Distribution-free comparison of monthly savings by group

Abstract

Background: An evaluation of a household financial coaching program. Methods: Monthly savings (usd) was compared between the coaching and waitlist groups with a two-sided Mann-Whitney U test (asymptotic method with continuity correction). Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

Introduction

We report an evaluation of a household financial coaching program. The primary question concerns participation in the financial coaching program and monthly savings (USD). The accompanying dataset (data.csv) contains one row per household-record as exported from the study database, including some duplicated exports and sentinel-coded missing values; it is described in the data dictionary.

Methods

Participants and data. The export contains 453 records. Variables are listed in the data dictionary. Group membership is recorded in `program`.

Data cleaning and exclusions. Exact duplicate records (identical on every column) were removed, keeping the first occurrence. The value -999 denotes a missing measurement and was treated as missing. Only households aged 18 to 80 years inclusive were analysed. Group labels in `program` were entered free-text and were normalised by trimming whitespace and lower-casing before use. Records with missing savings were excluded (complete-case analysis).

Statistical analysis. Monthly savings (usd) was compared between the coaching and waitlist groups with a two-sided Mann-Whitney U test (asymptotic method with continuity correction). The U statistic is reported for the coaching group. The rank-biserial correlation was computed as 1 - 2U/(n1 n2). Medians and the difference in medians (coaching minus waitlist) are reported. All analyses used the cleaned analysis sample; no random resampling was used.

Results

The coaching group ($n = 207$, median 377.90) differed from the waitlist group ($n = 204$, median 314.00); the difference in medians was 63.90 ($U = 28388.00$, $p < 0.001$, rank-biserial $r = -0.345$).

Discussion

The analysis followed the pre-specified plan. Limitations include reliance on database exports and complete-case handling of missing values.